KAIZEN MATHS

Lesson starter

Further Vectors: Lines, Planes and Products

Vector Lines

Name _____

SA1

Find the vector equation of each line.

Find a vector equation of the line through A = (2, -5, 6) and B = (4, -4, 4).

SA2

Determine whether each point lies on the given line.

Line L has equation $\mathbf{r} = \begin{pmatrix} 4 \\ -2 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 2 \\ 4 \end{pmatrix}$. Determine whether P = (8, -4, -10) lies on L.

Show your working

KAIZEN MATHS

Lesson starter

Further Vectors: Lines, Planes and Products

Vector Lines

Name _____

SB1

Find the point of intersection of each pair of vector lines.

Find the intersection of $L_1: \mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -6 \\ -5 \end{pmatrix}$ and $L_2: \mathbf{r} = \begin{pmatrix} -8 \\ 4 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 6 \\ 4 \end{pmatrix}$.

SB2

Find the shortest distance from each point to the given line.

Find the shortest distance from P = (-5, -4, 5) to the line $\mathbf{r} = \lambda \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Vector Lines

Name _____

EA1

Find the vector equation of each line.

Find a vector equation of the line through A = (6, -5, -2) and B = (11, 1, 2).

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Vector Lines

Name _____

EB1

Determine whether each point lies on the given line.

Line L has equation $\mathbf{r} = \begin{pmatrix} -5 \\ 6 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 5 \end{pmatrix}$.

Determine whether P = (-11, 9, 17) lies on L.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Vector Lines

Name _____

EC1

Find the point of intersection of each pair of vector lines.

Find the intersection of $L_1 : \mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -5 \\ 6 \end{pmatrix}$

and $L_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix}$.

Show your working

KAIZEN MATHS

Exit ticket

Further Vectors: Lines, Planes and Products

Vector Lines

Name _____

EA1

Find the vector equation of each line.

Find a vector equation of the line through A = (6, -5, -2) and B = (11, 1, 2).

Show your working

Teacher answers & working

Further Vectors: Lines, Planes and Products · Vector Lines

ALG-ADB4-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

SA1

Find the vector equation of each line.

Find a vector equation of the line through A = (2, -5, 6) and B = (4, -4, 4).

Answer:

$$\mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

Working:

A point on the line is (2, -5, 6).

Direction vector AB = $\begin{pmatrix} 4 \\ -4 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix}$

$$\mathrm{AB} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

Use $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.

$$\mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

SA2

Determine whether each point lies on the given line.

Line L has equation $\mathbf{r} = \begin{pmatrix} 4 \\ -2 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 2 \\ 4 \end{pmatrix}$. Determine whether P = (8, -4, -10) lies on L.

Answer:

Yes, the point lies on the line.

Working:

Compare components with $\begin{pmatrix} 8 \\ -4 \\ -10 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 2 \\ 4 \end{pmatrix}$.

x-component: 8 = 4 + -4λ

$\lambda = -1$

y-component: -4 = -2 + 2λ

$\lambda = -1$

z-component: -10 = -6 + 4λ

$\lambda = -1$

The x, y, and z components all give λ = -1.

Yes, the point lies on the line.

Teacher answers & working

Further Vectors: Lines, Planes and Products · Vector Lines

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SB1

Find the point of intersection of each pair of vector lines.

Find the intersection of L_1 : r = $\begin{pmatrix}1\\-2\\-6\end{pmatrix} + \lambda\begin{pmatrix}-3\\-6\\-5\end{pmatrix}$ and L_2 : r = $\begin{pmatrix}-8\\4\\-3\end{pmatrix} + \mu\begin{pmatrix}-3\\6\\4\end{pmatrix}$.

Answer:

Intersection = (-2, -8, -11)

Working:

At the intersection, the position vectors are equal.

$$\begin{pmatrix}1\\-2\\-6\end{pmatrix} + \lambda\begin{pmatrix}-3\\-6\\-5\end{pmatrix} = \begin{pmatrix}-8\\4\\-3\end{pmatrix} + \mu\begin{pmatrix}-3\\6\\4\end{pmatrix}$$

Equate components and solve for the two parameters.

A solution is $\lambda = 1$, $\mu = -2$.

Substitute $\lambda = 1$ into L_1.

$$\begin{pmatrix}1\\-2\\-6\end{pmatrix} + 1\begin{pmatrix}-3\\-6\\-5\end{pmatrix} = \begin{pmatrix}-2\\-8\\-11\end{pmatrix}$$

Intersection = (-2, -8, -11)

SB2

Find the shortest distance from each point to the given line.

Find the shortest distance from P = (-5, -4, 5) to the line r = $\lambda\begin{pmatrix}2\\1\\2\end{pmatrix}$.

Answer:

distance = $\frac{\sqrt{578}}{\sqrt{9}} \approx 8.014$

Working:

Use distance = $\frac{|AP \times d|}{|d|}$. Here A is the origin, so AP = $\begin{pmatrix}-5\\-4\\5\end{pmatrix}$.

$$\mathrm{AP} \times \mathrm{d} = \begin{pmatrix}-13\\20\\3\end{pmatrix}$$

$|\mathrm{AP} \times \mathrm{d}| = \sqrt{578}$

$|\mathrm{d}| = \sqrt{9}$

distance = $\frac{\sqrt{578}}{\sqrt{9}} \approx 8.014$

Teacher answers & working

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EA1

Find the vector equation of each line.

Find a vector equation of the line through A = (6, -5, -2) and B = (11, 1, 2).

Answer:

$$\mathbf{r} = \begin{pmatrix} 6 \\ -5 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 6 \\ 4 \end{pmatrix}$$

Working:

A point on the line is (6, -5, -2).

Direction vector AB = $\begin{pmatrix} 11 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 6 \\ -5 \\ -2 \end{pmatrix}$

$$\mathrm{AB} = \begin{pmatrix} 5 \\ 6 \\ 4 \end{pmatrix}$$

Use $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.

$$\mathbf{r} = \begin{pmatrix} 6 \\ -5 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 6 \\ 4 \end{pmatrix}$$

EB1

Determine whether each point lies on the given line.

Line L has equation $\mathbf{r} = \begin{pmatrix} -5 \\ 6 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 5 \end{pmatrix}$. Determine whether P = (-11, 9, 17) lies on L.

Answer:

Yes, the point lies on the line.

Working:

Compare components with $\begin{pmatrix} -11 \\ 9 \\ 17 \end{pmatrix} = \begin{pmatrix} -5 \\ 6 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 5 \end{pmatrix}$.

x-component: -11 = -5 + -2λ

$\lambda = 3$

y-component: 9 = 6 + 1λ

$\lambda = 3$

z-component: 17 = 2 + 5λ

$\lambda = 3$

The x, y, and z components all give λ = 3.

Yes, the point lies on the line.

Teacher answers & working

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EC1

Find the point of intersection of each pair of vector lines.

Find the intersection of L_1 : r = $\begin{pmatrix} 5 \\ -2 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -5 \\ 6 \end{pmatrix}$ and L_2 : r = $\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix}$.

Answer:

Intersection = (7, 3, -12)

Working:

At the intersection, the position vectors are equal.

$$\begin{pmatrix} 5 \\ -2 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -5 \\ 6 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix}$$

Equate components and solve for the two parameters.

A solution is $\lambda = -1$, $\mu = -2$.

Substitute $\lambda = -1$ into L_1.

$$\begin{pmatrix} 5 \\ -2 \\ -6 \end{pmatrix} + -1 \begin{pmatrix} -2 \\ -5 \\ 6 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \\ -12 \end{pmatrix}$$

Intersection = (7, 3, -12)